EXP. NO. 5 DATE: 02/09/2025

CONFIGURE IPS/IPS IN CISCO PACKET TRACER

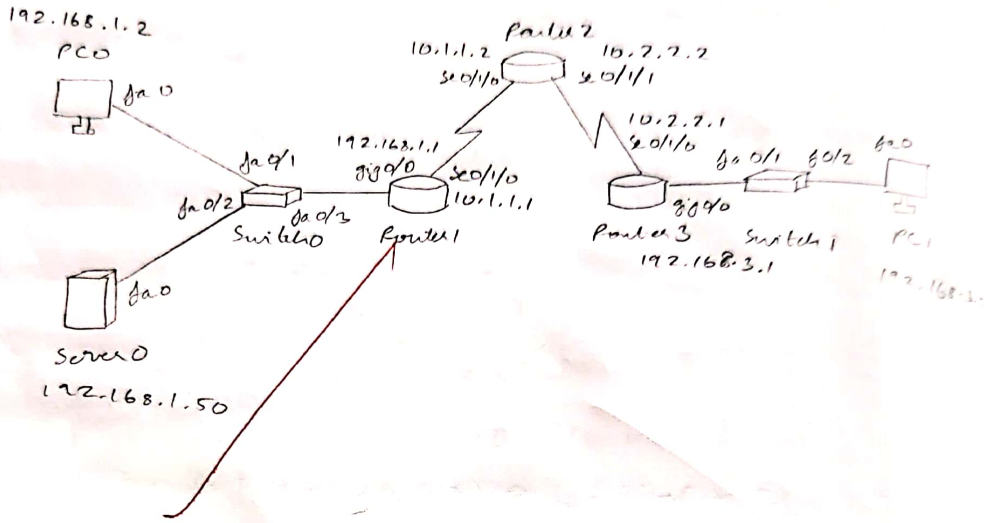
AIM

To configure IOS IPS on R1 in Cisco Packet Tracer to inspect traffic headed into the 192.168.1.0/24 LAN, log events to the Syslog server (192.168.1.50) with accurate timestamps.

PROCEDURE

1. Open the given topology with R1-R2-R3 connected by serial links and LANs 192.168.1.0/24 and 192.168.3.0/24 as per the addressing table.
2. On R1, enable local user authentication and apply it to console access so only authenticated users can reach EXEC/privileged modes.
3. Activate the security feature set on R1.
4. Verify baseline connectivity: ping between PC-A and PC-C both ways.
5. On R1, create a storage directory in flash for IPS files.
6. Point the IPS configuration to that directory, create an IPS policy named "iosips", and enable IPS logging.
7. Set the correct date and time on R1 and enable logging timestamps with precision.
8. Configure R1 to send log messages to the Syslog server at 192.168.1.50.

TOPOLOGY


192.168.1.2
PC0
fa0
fa0/1
fa0/2
fa0/3
Switch0
fa0
Server0
192.168.1.50
192.168.1.1
gig0/0
se0/1/0
10.1.1.1
Router1
Router2
10.1.1.2
se0/1/0
10.2.2.2
se0/1/1
10.2.2.1
se0/1/0
gig0/0
Router3
192.168.3.1
fa0/1
fa0/2
Switch1
fa0
PC1
192.168.3.

OUTPUT

(i) Initial connectivity test (Before IPS) :-

From PC-A (192.168.1.2) ping PC-C (192.168.3.2):

PC> ping 192.168.3.2

→ ping successful

From PC-C (192.168.3.2) ping PC-A (192.168.1.2):

PC> ping 192.168.1.2

→ ping successful

(ii) Connectivity test after IPS enabled :-

From PC-C (192.168.3.2) ping PC-A (192.168.1.2):

PC> ping 192.168.1.2

→ ping failed

From PC-A (192.168.1.2) ping PC-C (192.168.3.2):

PC> ping 192.168.3.2

→ ping successful

(iii) Syslog server:

Jul 9 19:26:02.345: %IPS-6-ALERT:
sig:2004 subsig:0 ICMP Echo Request

Action: deny-packet-inline

Src: 192.168.3.2 Dst: 192.168.1.2

(iv) IPS Statistics:

R1# show ip ips statistics

IPS statistics

Packets scanned: 20

Packets dropped: 4

Alerts sent: 4

9. In the IPS signature categories on R1, retire all signatures to start clean.

10. Unretire the "ios-ips basic" category so only the basic signatures to start clean.

11. Apply IPS policy to interface G0/1 in the outbound direction so traffic destined to the 192.168.1.0/24 LAN is inspected as it leaves R1 toward that LAN.

12. Edit signature ID 2004, subsig 0 (ICMP Echo), mark it as not retired and enabled, and set its actions to produce an alert and to deny the packet inline.

13. End configuration and save the running configuration to startup.

14. From PC-C, ping PC-A, observe behaviour.

15. From PC-A, ping PC-C, observe behaviour.

16. On the Syslog server, open the log viewer and check for IPS alerts corresponding to the ICMP traffic burst.

RESULT

Thus, IPS was successfully deployed on R1 to protect the 192.168.1.0/24 network.